

Assignment 1: Mathematical Foundations

ME/BME 736 — Advanced Finite Element Methods

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Name: _____

Date: _____

Show the main steps in each calculation. When a requested maximum or minimum does not exist, state why it does not exist.

Problem 1 (10 marks): Sequences and limits

1. Determine the limit of

$$x_n = \left(\frac{1}{n}, 1 - \frac{1}{n} \right)$$

in $\mathbb{R}^2$.

2. Show that

$$x_n = (-1)^n + \frac{1}{n}$$

does not converge, and identify two convergent subsequences. State the limit of each subsequence.

Problem 2 (20 marks): Bounds and extrema

1. For each set, determine sup, inf, max, and min when they exist:

$$(2, 5), \quad [2, 5), \quad \left\{ 1 - \frac{1}{n} : n \geq 1 \right\}, \quad \{(-1)^n : n \geq 1\}.$$

2. Let $f(x) = x(1 - x)$. Determine the supremum, infimum, maximum, and minimum of f on each of the following domains:
- $(0, 1)$;
 - $[0, 1]$.
3. Let $f(x) = 1/x$ on $(0, 1)$.

- a. Is f bounded above?
- b. Does f have a minimum?

Problem 3 (10 marks): Cartesian vectors

In the Cartesian basis of $\mathbb{R}^3$, let

$$\mathbf{a} = 2\mathbf{e}_1 - \mathbf{e}_2 + 2\mathbf{e}_3, \quad \mathbf{b} = \mathbf{e}_1 + 3\mathbf{e}_2.$$

1. List the components of $\mathbf{a}$ and $\mathbf{b}$.
2. Compute $\mathbf{a} \cdot \mathbf{b}$, $\|\mathbf{a}\|$, and $\|\mathbf{b}\|$.
3. Use $a_i = \mathbf{a} \cdot \mathbf{e}_i$ to recover each component of $\mathbf{a}$.

Problem 4 (10 marks): Tensor-vector multiplication

Let $\mathbf{c} = \mathbf{A}\mathbf{b}$. Starting from the basis expansions

$$\mathbf{A} = A_{ij}\mathbf{e}_i \otimes \mathbf{e}_j, \quad \mathbf{b} = b_k\mathbf{e}_k,$$

use $\mathbf{e}_j \cdot \mathbf{e}_k = \delta_{jk}$ and the defining property of the Kronecker delta to show that

$$c_i = A_{ij}b_j.$$

Problem 5 (50 marks): Identifying real vector spaces

For each set, use the usual addition and multiplication by scalars from $\mathbb{R}$. Determine whether these operations make the set a real vector space. If they do not, identify at least one vector-space requirement that fails.

1. $\mathbb{Q}$, the rational numbers;
2. $C(\overline{\Omega}) = \{u : \overline{\Omega} \rightarrow \mathbb{R} : u \text{ is continuous}\}$, with operations defined pointwise;
3. $\{u \in C(\overline{\Omega}) : u(\mathbf{x}_0) = 1\}$, where $\mathbf{x}_0 \in \overline{\Omega}$ is fixed;
4. $\{(x, y) \in \mathbb{R}^2 : y = x\}$;
5. $\{(x, y) \in \mathbb{R}^2 : y = x + 1\}$.